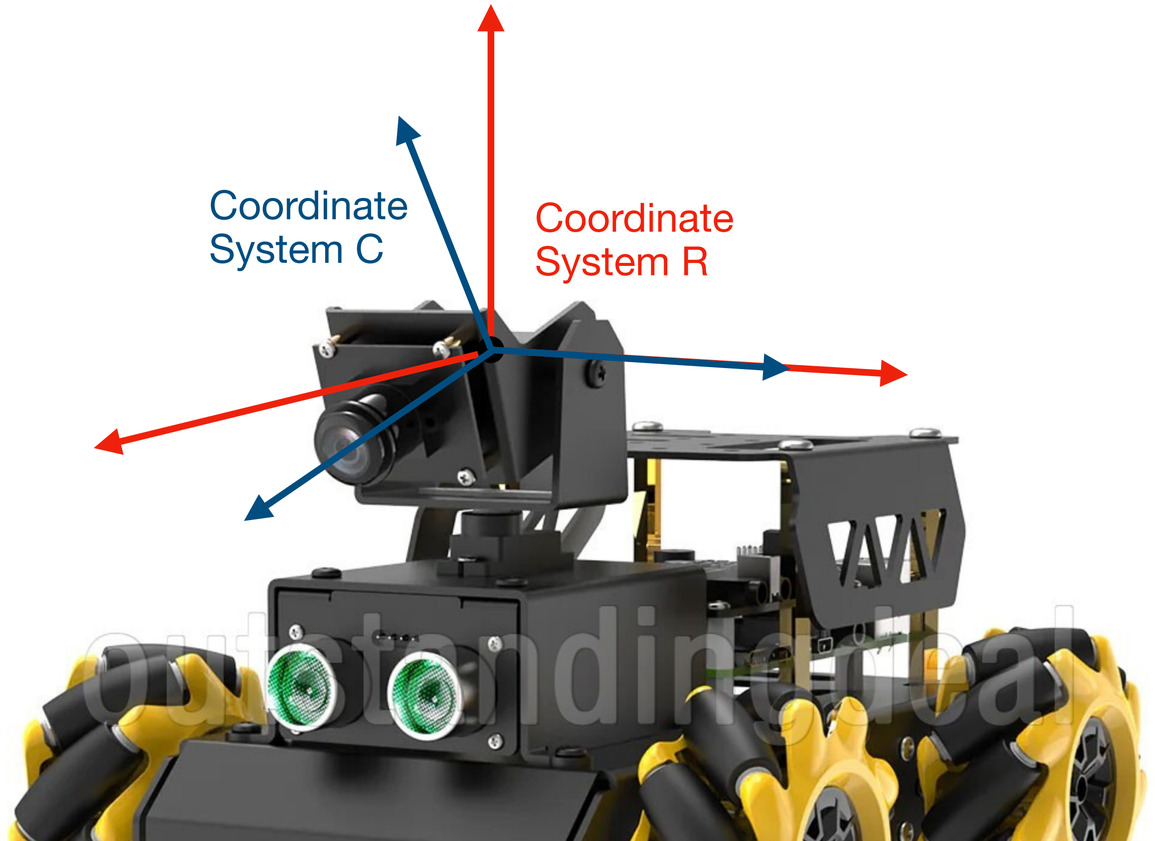
# ROB-UY 3303

**Robot Motion & Planning**

# Lecture 02B - Robot Represenetations



## Agenda

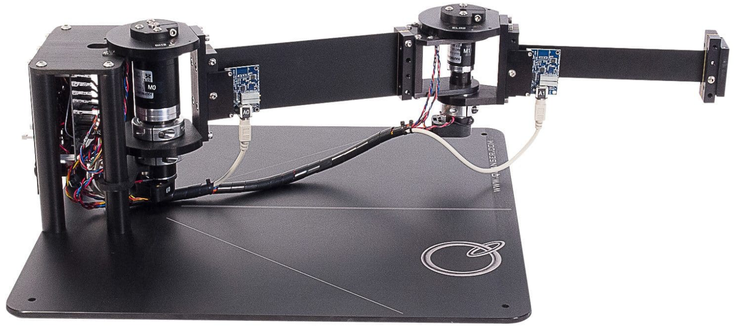
1. Degrees of Freedom
2. Configuration Space for Planning
3. Coordinate Frames

## Agenda

1. **Degrees of Freedom**
2. Configuration Space for Planning
3. Coordinate Frames

**Configuration:**

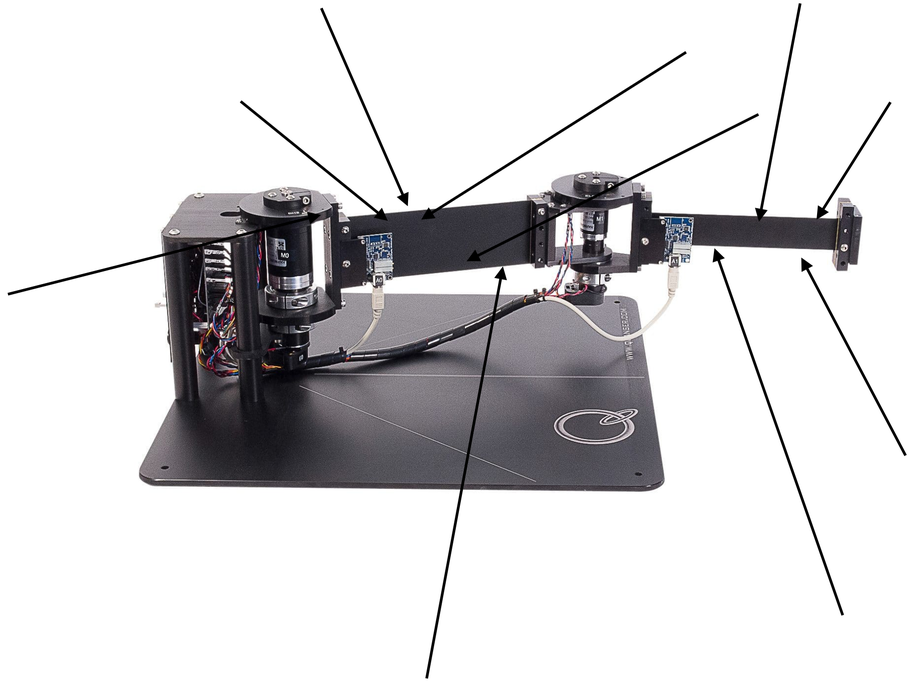
- The specification of the positions of all points of a mechanism



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**Configuration:**

- The specification of the positions of all points of a mechanism

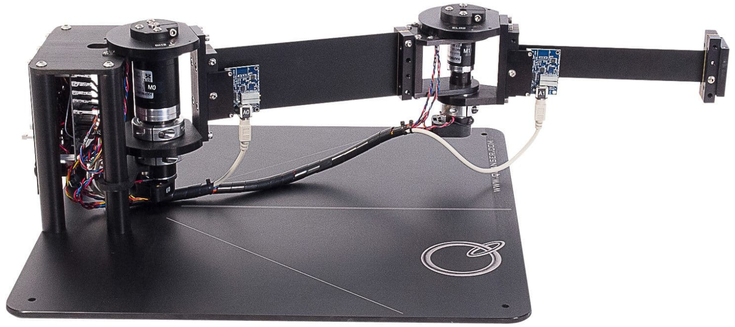


**Configuration:**

- The specification of the positions of all points of a mechanism

**Degrees of Freedom:**

- Number of real numbers required to describe a configuration

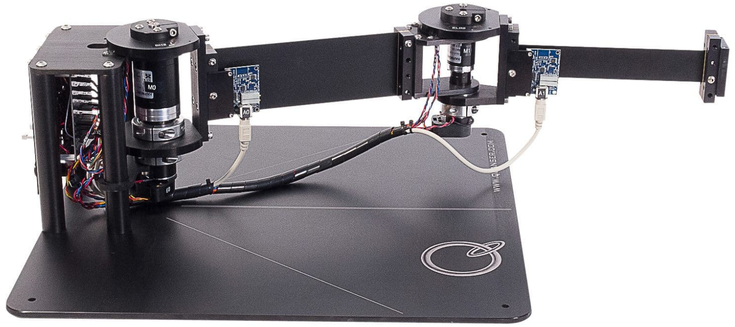


**Configuration:**

- The specification of the positions of all points of a mechanism

**Degrees of Freedom:**

- Number of real numbers required to describe a configuration



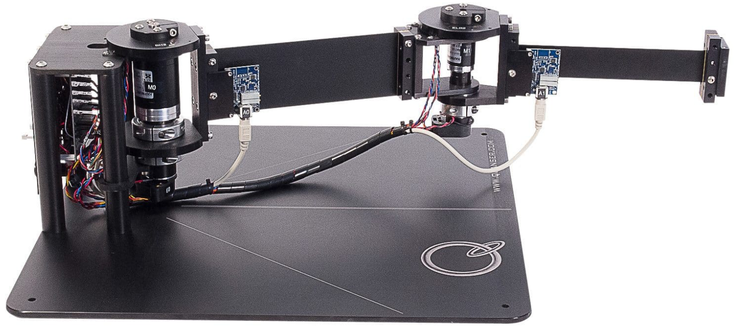
nDOF = 2

**Configuration:**

- The specification of the positions of all points of a mechanism

**Degrees of Freedom:**

- Number of real numbers required to describe a configuration



**Configuration Space:**

- The nDOF dimension space of all configurations

**Configuration:**

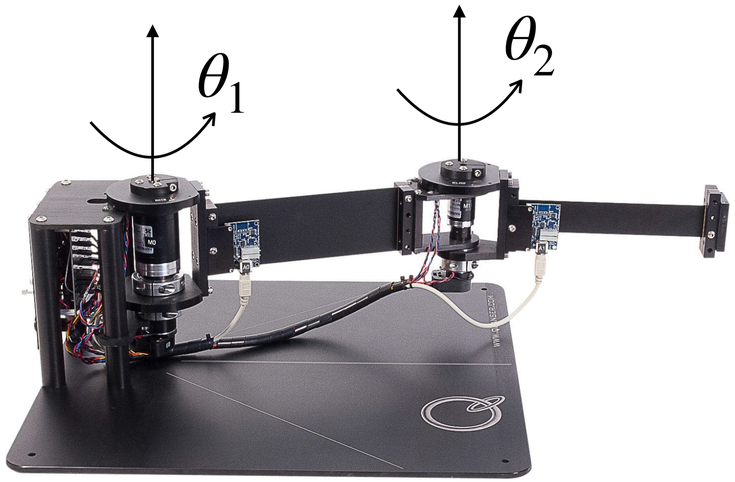
- The specification of the positions of all points of a mechanism

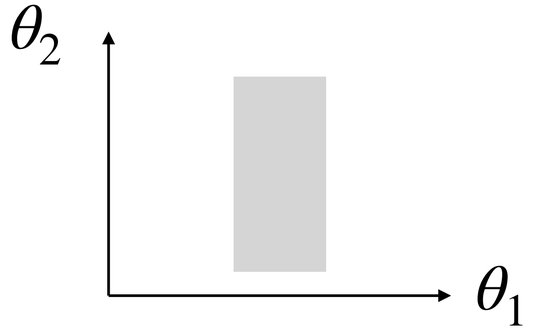
**Degrees of Freedom:**

- Number of real numbers required to describe a configuration

**Configuration Space:**

- The nDOF dimension space of all configurations



**Planar (Rigid) Body:**

- The nDOF m is 3



**Spatial (Rigid) Body:**

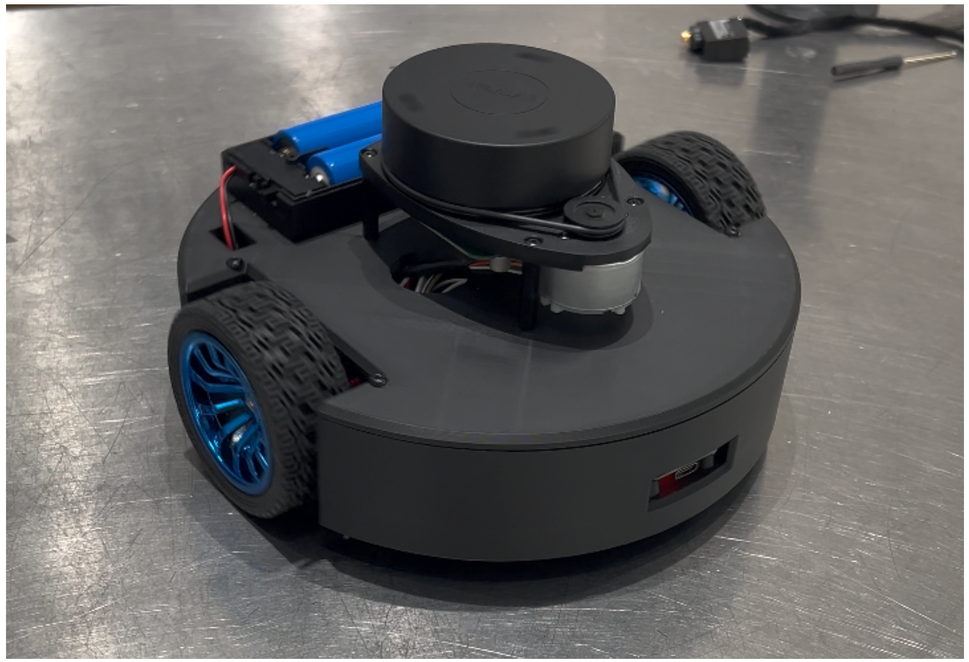
- The nDOF m is 6



NYU

**Differential Drive Robot:**

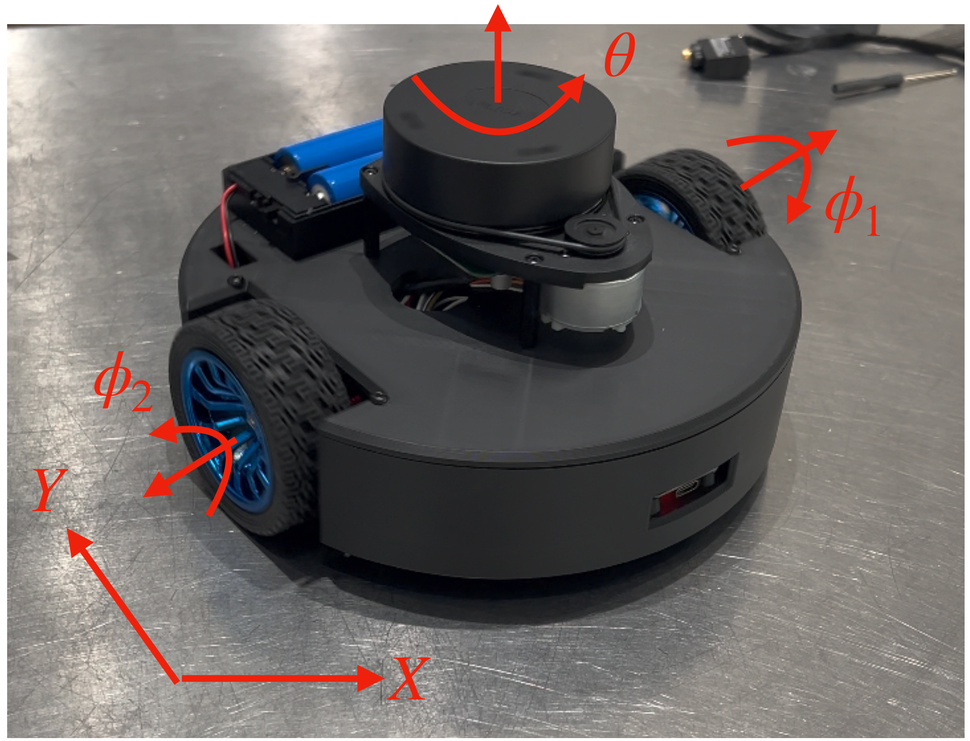
- The nDOF m is …



**Differential Drive Robot:**

- The nDOF m is 5

$$[X, Y, \theta, \phi_1, \phi_2]$$



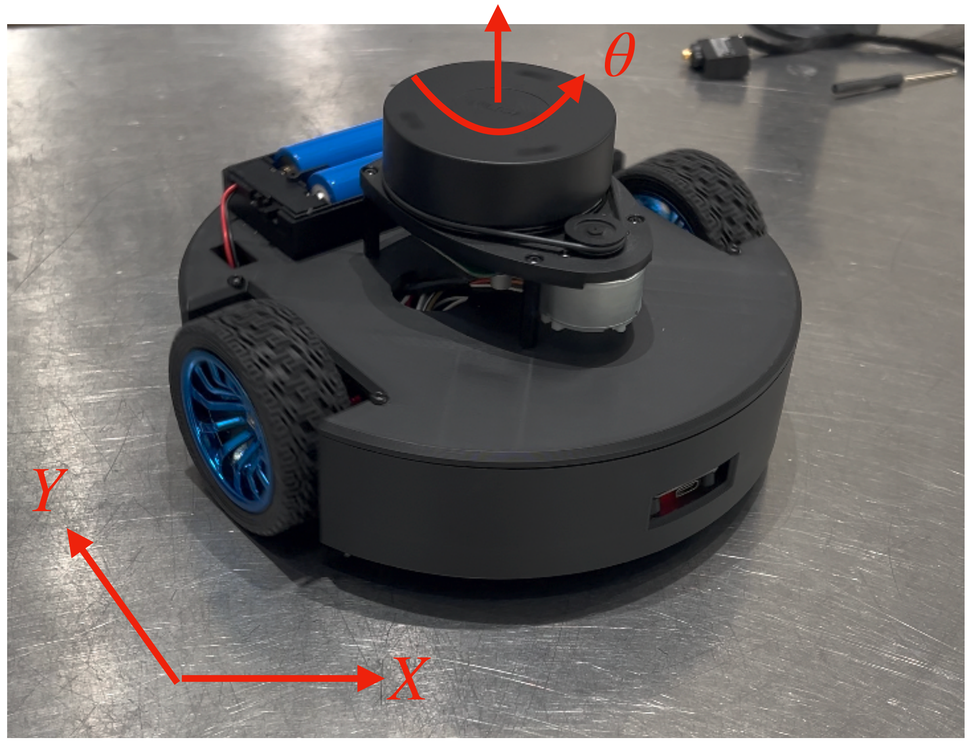
**Differential Drive Robot:**

- The nDOF m is 5

$$[X, Y, \theta, \phi_1, \phi_2]$$

- We may plan in a space of 3

$$[X, Y, \theta]$$

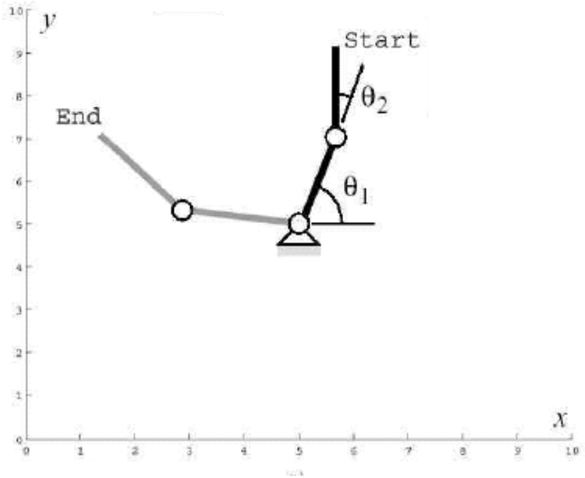


## Agenda

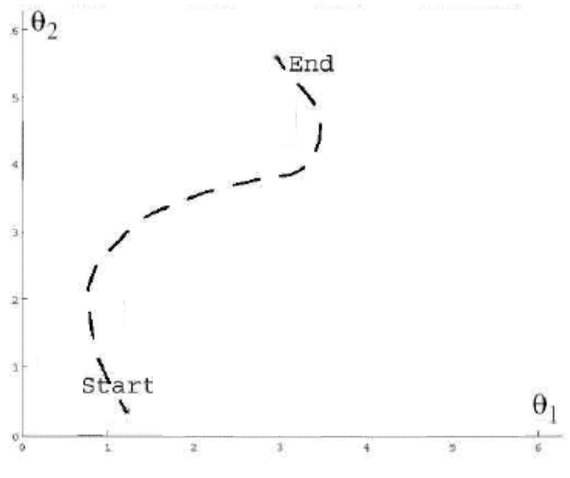
1. Degrees of Freedom
2. **Configuration Space for Planning**
3. Coordinate Frames

# Configuration Space For Planning

- A **configuration** $q$ will completely define the position of every point on a robot
- The **configuration space** $C$, is the space of all possible configurations of the robot.
- The **free space** $F \in C$, is the portion of the configuration space which is collision-free.



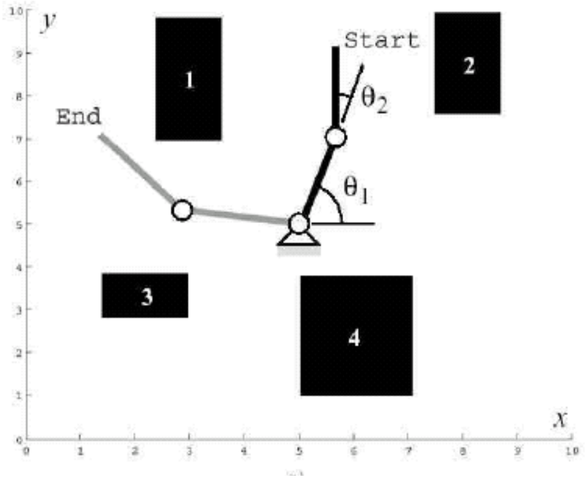

Workspace



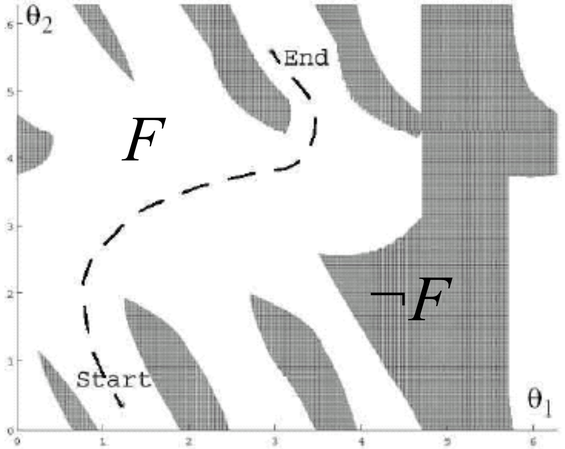

Configuration Space

# Configuration Space For Planning

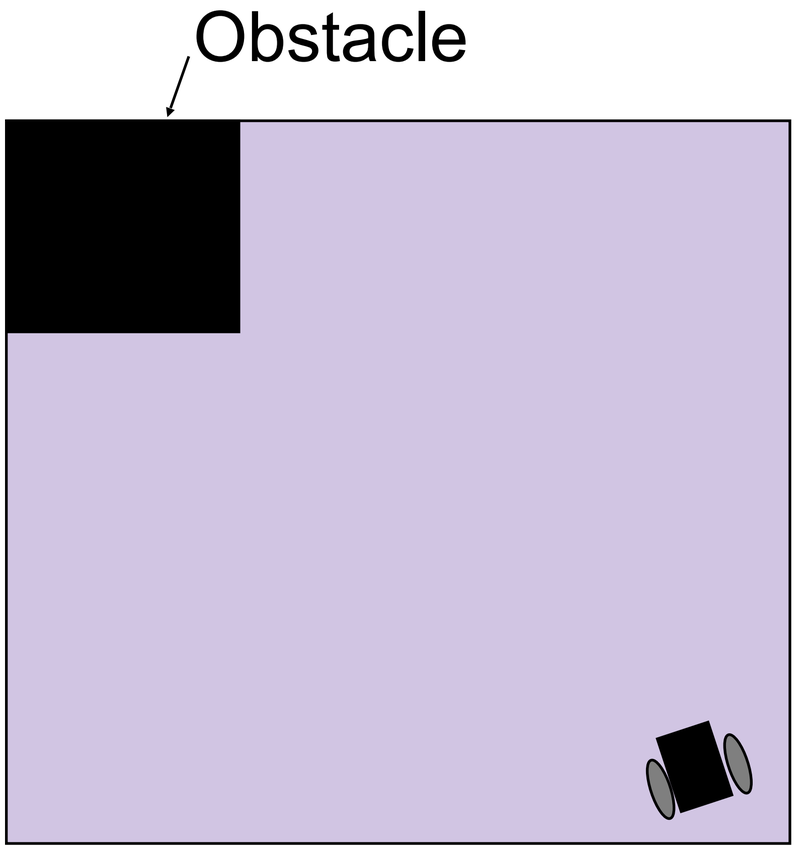
- A **configuration** $q$ will completely define the state of a robot (e.g. mobile robot $x, y, \theta$)
- The **configuration space** $C$, is the space of all possible configurations of the robot.
- The **free space** $F \in C$, is the portion of the free space which is collision-free.



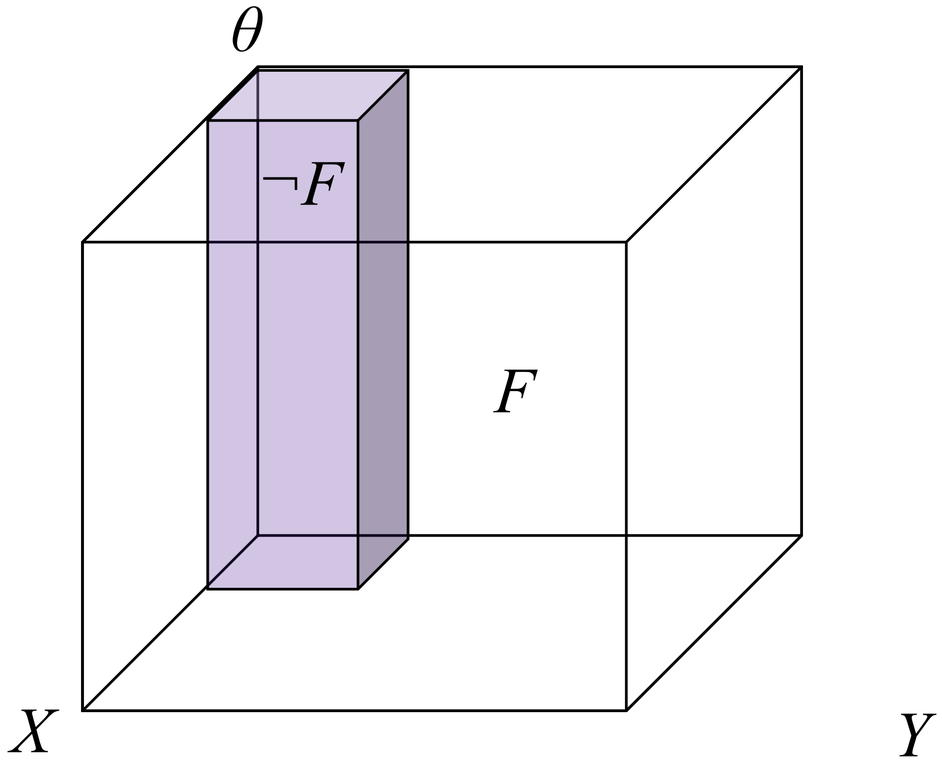

Workspace



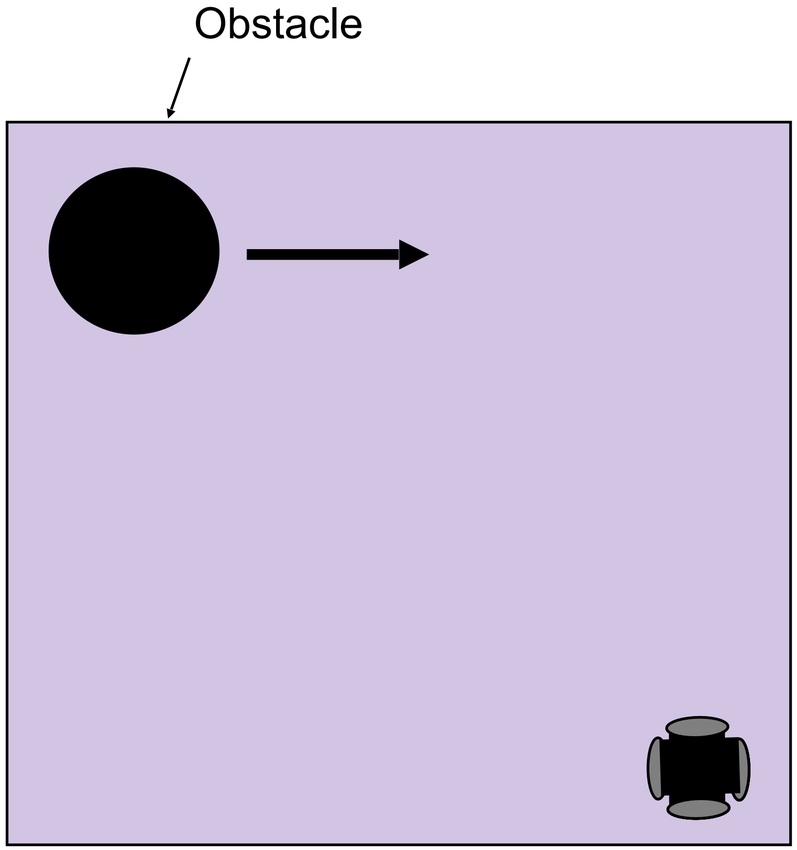

Configuration Space



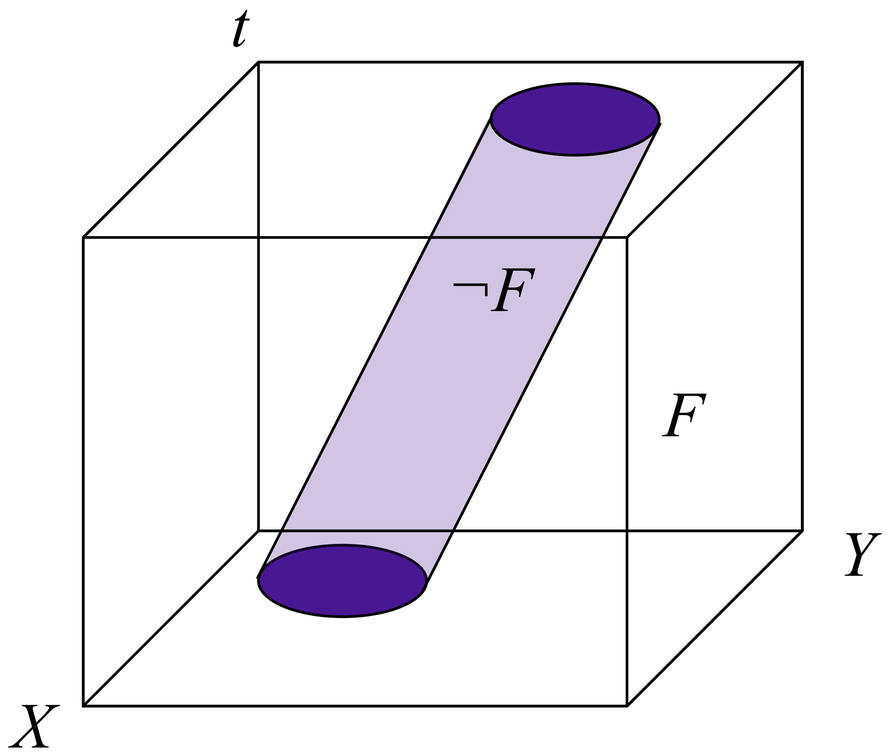

Workspace




Configuration Space




Workspace



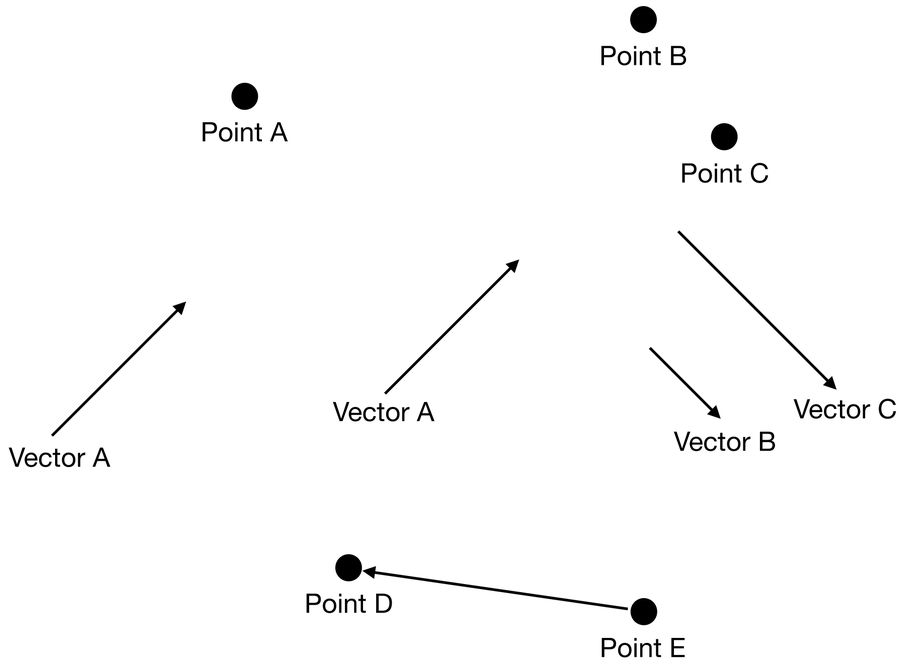

Configuration Space

## Agenda

1. Degrees of Freedom
2. Configuration Space for Planning
3. **Coordinate Frames**

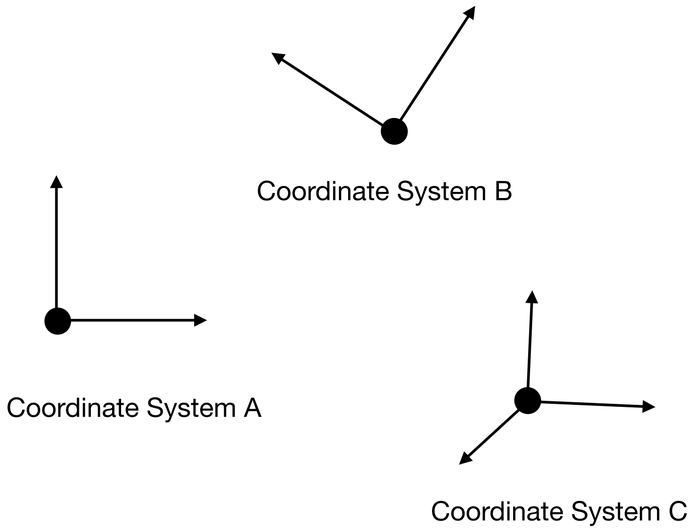
## Points & Vectors

- A **point** defines a specific **position in space**
- A **vector** is combination of a **direction and magnitude**
  - A vector can be used to define the position of one point in space relative to another.



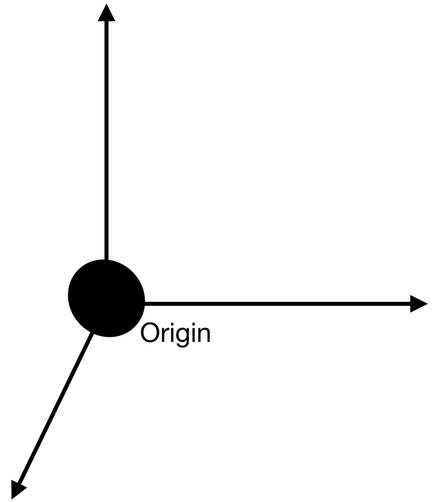
# Coordinate Systems

- AKA a **Reference Frame**
- Defined by 1 point and 2 vectors for 2D spaces
- Defined by 1 point and 3 vectors for 3D spaces



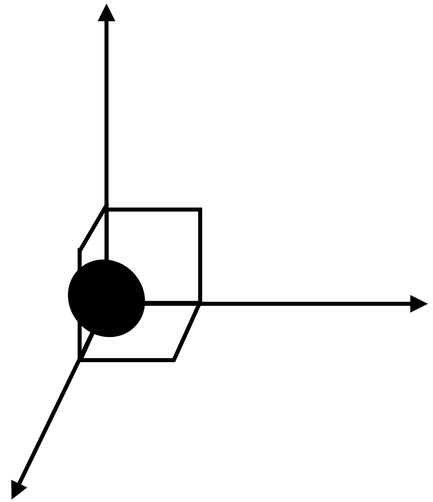
## Coordinate Systems

- The point is called the origin



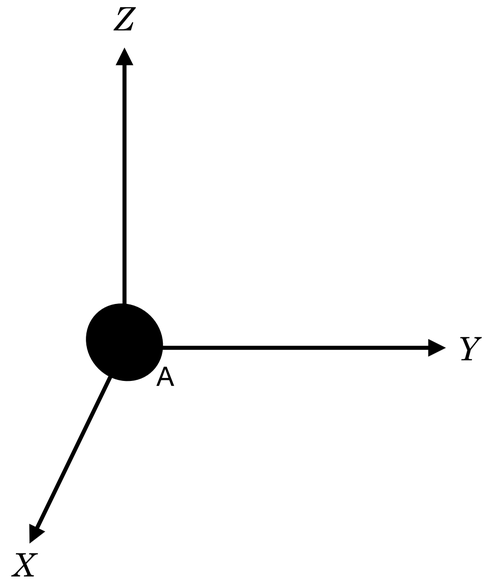
## Coordinate Systems

- The point is called the origin
- The vectors have perpendicular directions



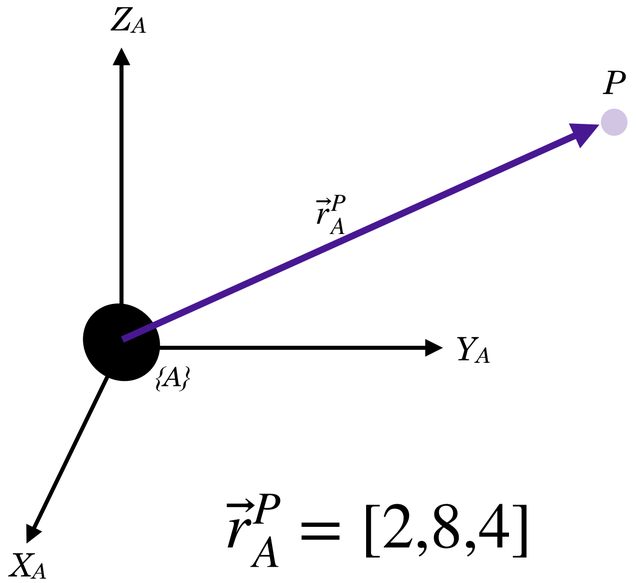
## Coordinate Systems

- The point is called the origin
- The vectors have perpendicular directions
- Each direction vector is assigned a label



## Coordinate Systems

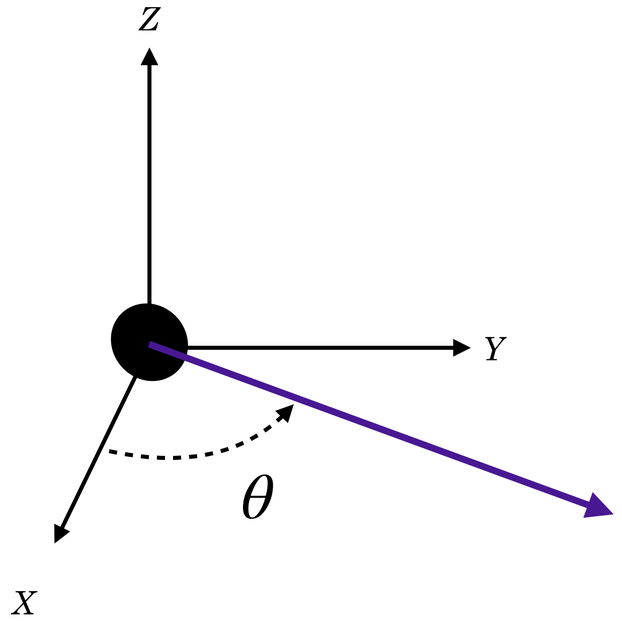
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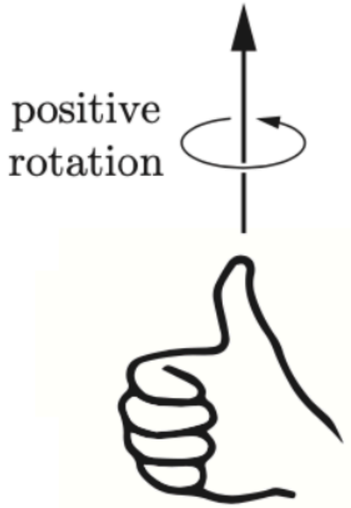



$$\vec{r}_A^P = [2,8,4]$$

## Coordinate Systems

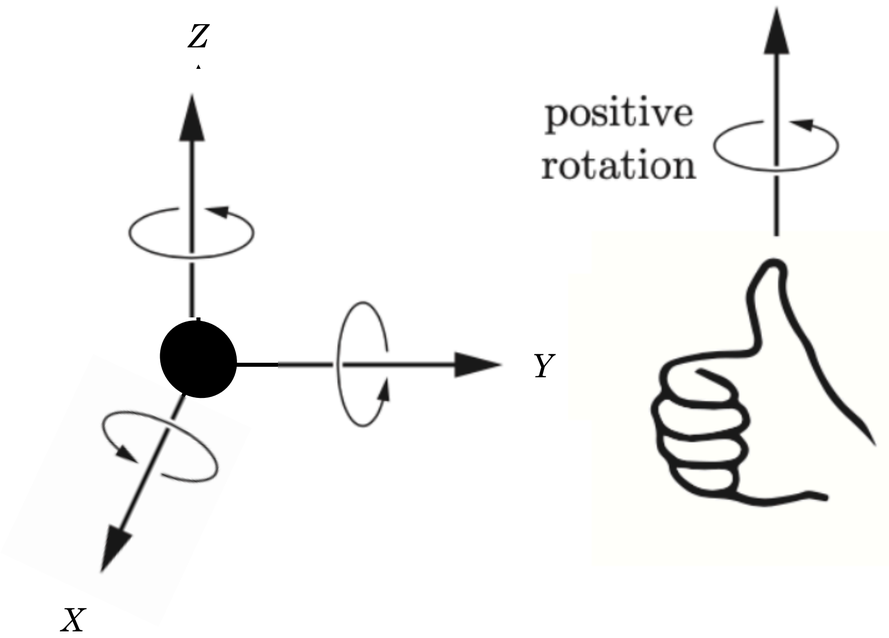
- The point is called the origin
- The vectors have perpendicular directions
- Each direction vector is assigned a label
- **Positive rotations** are CCW when looking in the in the negative direction of the axis of rotation.



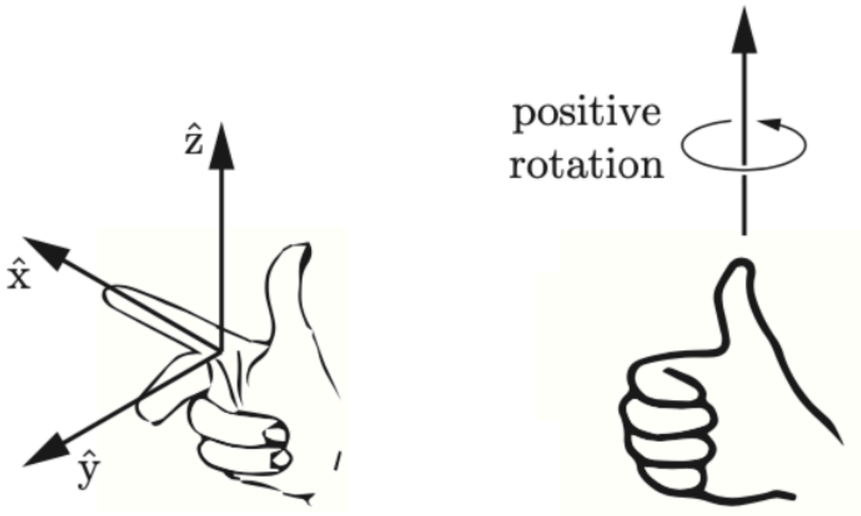
## Coordinate Systems

- The point is called the origin
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- **Positive rotations** are CCW when looking in the in the negative direction of the axis of rotation.



## Coordinate System Conventions in Robotics

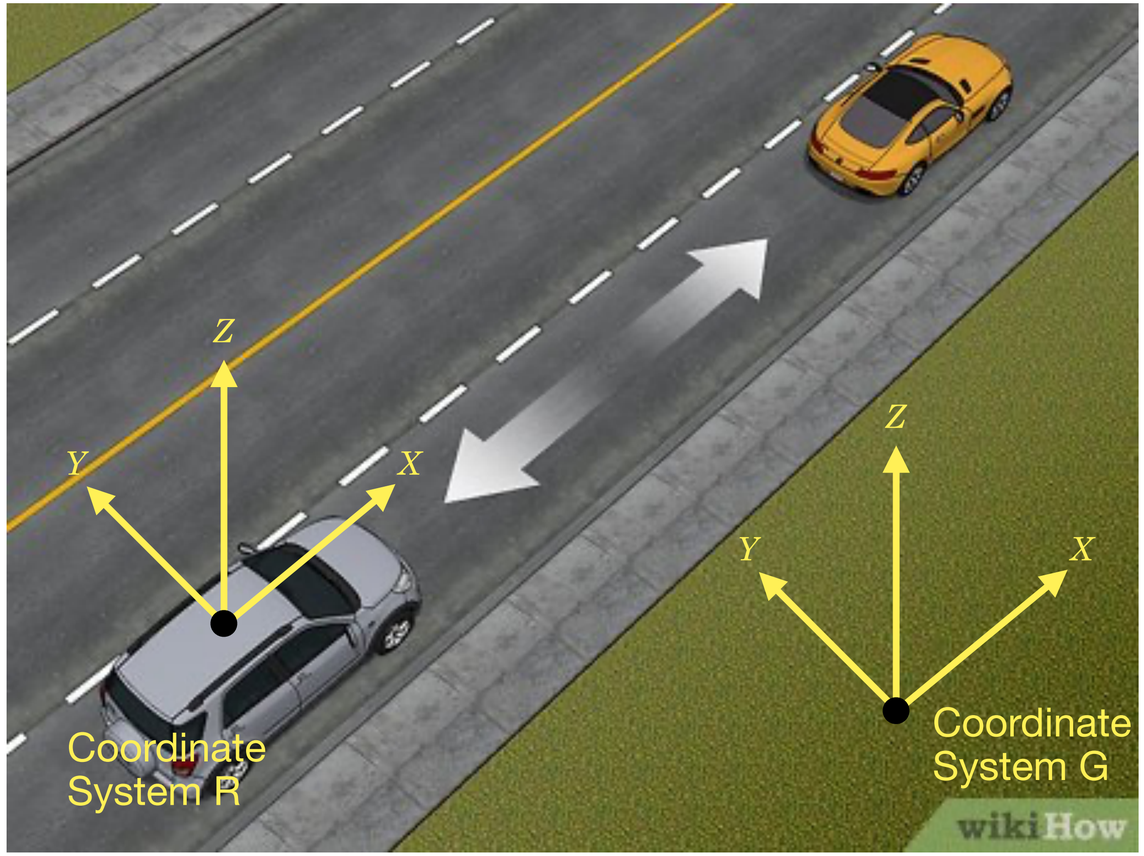
- **Right hand coordinate frames** have
  - Rotating $x$-axis about the $z$-axis yields the $y$-axis
  - Rotating $y$-axis about the $x$-axis yields the $z$-axis
  - Rotating $z$-axis about the $y$-axis yields the $x$-axis




Right-handed frame

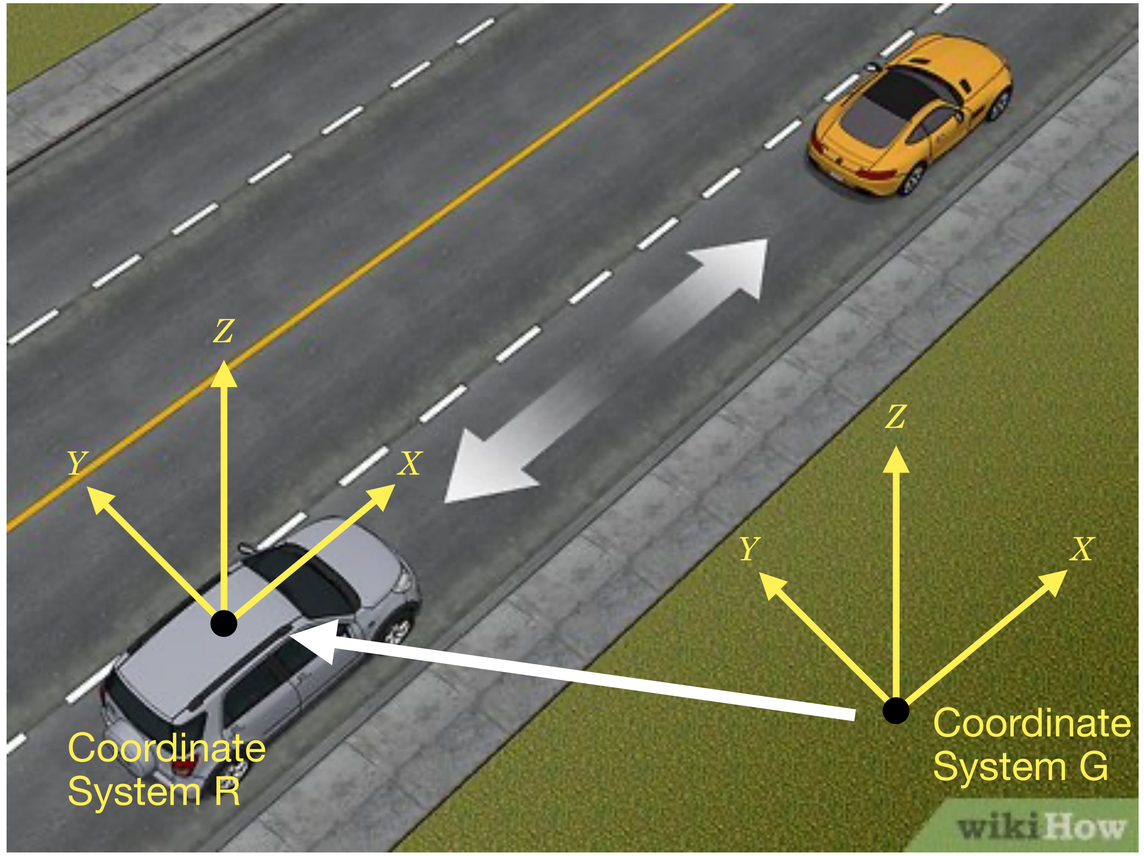
## Coordinate Systems

- In robotics, the frame is attached to something physical (e.g. a robot or the ground).



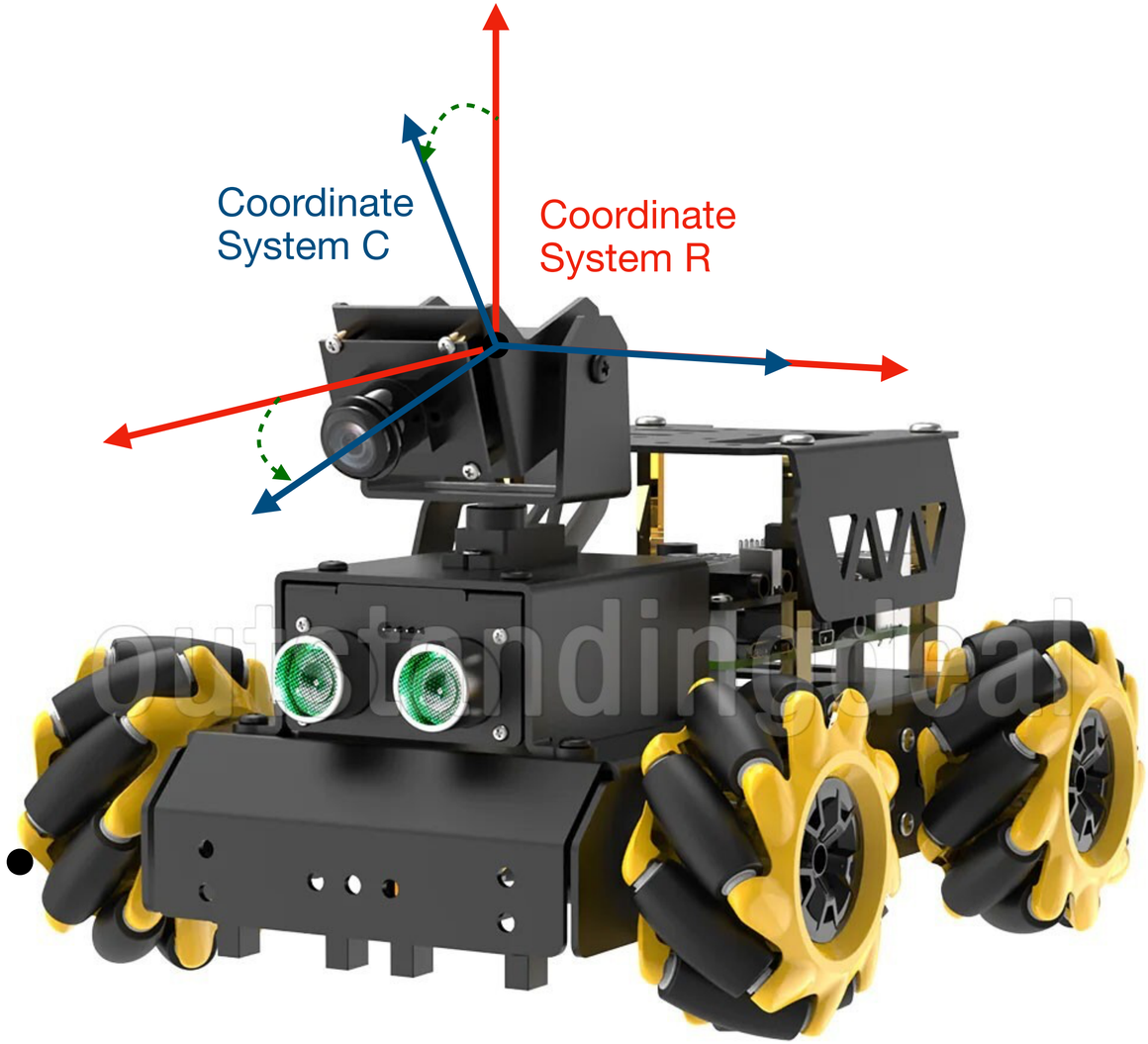
## Coordinate Systems

- In robotics, the frame is attached to something physical (e.g. a robot or the ground).
- It is helpful to be able to:
  - Describe the **relative position** of one frame with respect to another frame



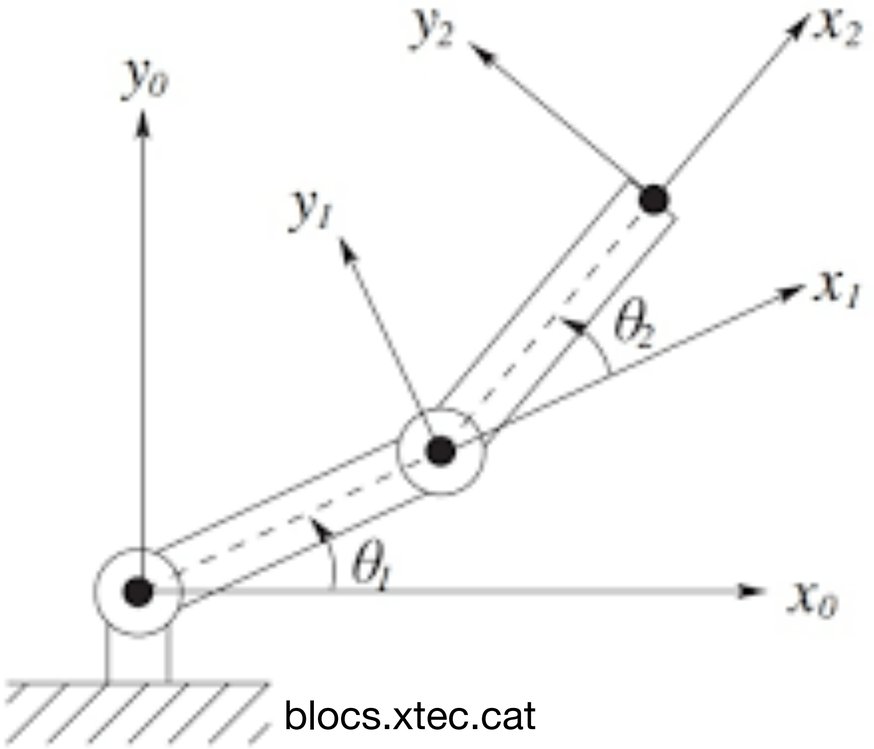
## Coordinate Systems

- In robotics, the frame is attached to something physical (e.g. a robot or the ground).
- It is helpful to be able to:
  - Describe the relative position of one frame with respect to another frame
  - Describe the **relative rotation** of one frame with respect to another frame



## Coordinate Systems

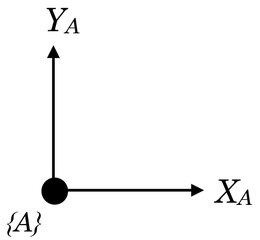
- In robotics, the frame is attached to something physical (e.g. a robot or the ground).
- It is helpful to be able to:
  - Describe the **relative position** and **relative rotation** of one frame with respect to another frame

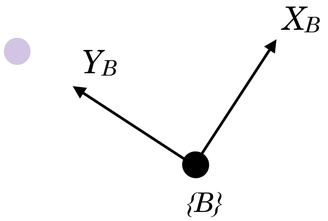



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## Coordinate Systems

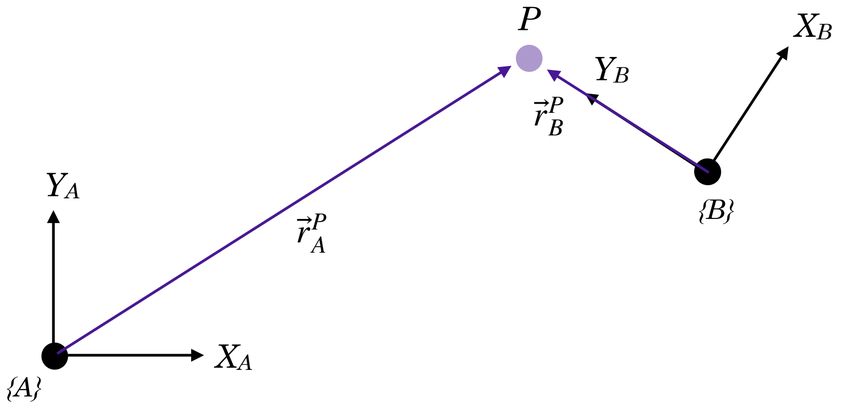
- It is important to note that a point in space looks very different from the perspective of one frame versus another.



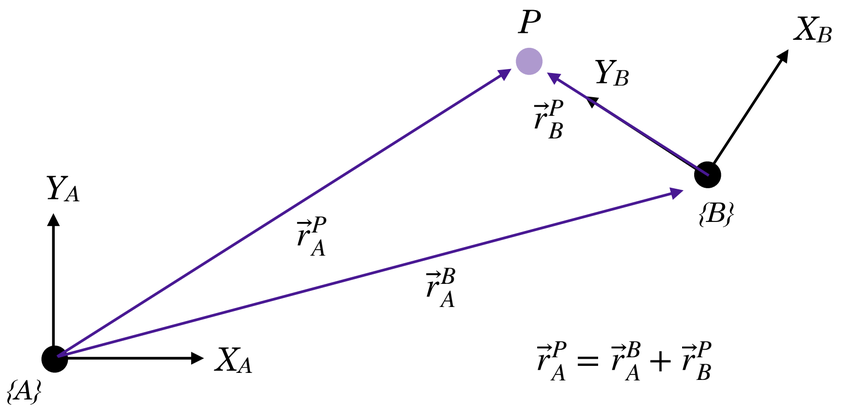
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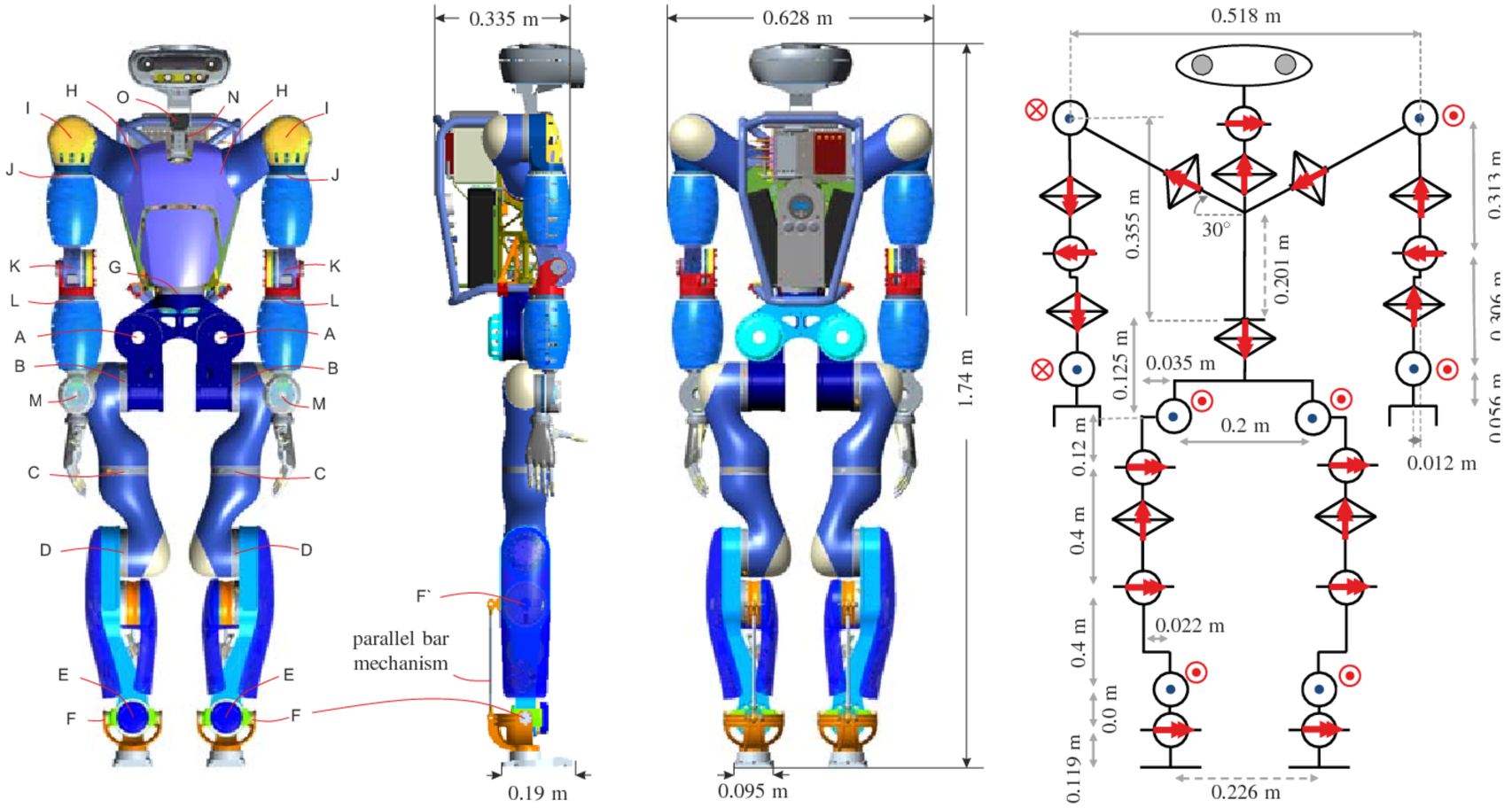
## Coordinate Systems

- It is important to note that a point in space looks very different from the perspective of one frame versus another.



# ROB-UY 3303

## Robot Motion & Planning




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